

One button takes a night-shift supervisor from line stoppage to root cause, with evidence anyone can check.

重點 給中小代工廠夜班主管的停線調查 Copilot · 一鍵找根因 · 每個結論都查得到證據。

PROBLEM

3 a.m., Line 2 stops. The night supervisor is alone; engineers are off site. Finding the cause means digging through Excel sheets, PLC CSV exports and shift logs, about 40 minutes [pending interviews], while every hour of downtime costs X [pending interviews].

重點 凌晨停線、主管一個人翻 Excel 找原因。40 分鐘和每小時損失都還沒有訪談數字。

SOLUTION

One **Investigate** button, no chat box. Gemini calls five fixed, tested queries and never writes SQL (demo data on DuckDB, swappable for BigQuery). Every step becomes an evidence card that opens to its source rows. No evidence, no conclusion: it answers “Insufficient evidence” instead of guessing.

1 Alarm

03:00, Line 2 turns red; the supervisor presses Investigate.

2 Evidence

Cards appear one by one (temperature, coolant flow, valve, shift log), each tracing back to source data rows.

3 Action

Root cause “cooling valve CV-2 stuck” with a server-scored confidence label; one click creates a work order, opened on a phone by QR.

重點 Gemini 只能選固定查詢，數字可回查原始列，證據不足就出灰卡。實測約 12 秒找到根因（中位數 12.6 秒 · gemini-2.5-flash · docs/qa/runs）。

WHY NOW, WHY JAPAC SMBS

- “China plus one” is moving production to Vietnam, Thailand and Malaysia [to verify].
- Target: contract manufacturers with 50–300 staff [pending interviews], no MES, no data team.
- Gemini function calling makes a constrained, auditable agent practical on files these plants already export.

重點 產能南移（來源待查證），目標客戶沒有 MES 也沒有資料團隊。

BUSINESS MODEL

- Onboarding fee:** X one-time, to turn Excel and PLC exports into clean data [pending interviews]
- Subscription:** X per production line per month [pending interviews]
- Priced below the downtime and labour it saves; customer payback in X months [pending interviews]

重點 導入費加每條產線月訂閱；金額、回本月數都要等訪談。

DIFFERENTIATION

	Built for	Starts from
Siemens Industrial Copilot	Large plants in the Siemens automation ecosystem to verify	Automation and MES data to verify
Cognite	Asset-heavy enterprises with data teams to verify	Industrial data platform to verify
Google MDE	Plants building a connected data platform on Google Cloud to verify	Connected machine data to verify
LineSleuth	Small contract manufacturers without MES	Excel sheets and PLC CSV exports

重點 跟大廠服務的不是同一群客戶；Google MDE 是客戶以後的升級路徑，不當對手講。

BUILT ON GOOGLE CLOUD

Cloud Run · Singapore

DuckDB · demo data, swappable for BigQuery

Vertex AI Gemini · function calling

Cloud Logging · replayableIAM · least privilege

Budget alerts + rate limits

TRACTION AND VALIDATION

- Prototype on simulated data: one stoppage and one healthy-data scenario
- Regression set: 10 known root causes; score 10/10 on real Gemini, 3 runs each (bar 9/10)
- Time to root cause: ~12 s (measured median 12.6 s, docs/qa/runs)
- Plant-manager interviews: [pending interviews]

TEAM AND CONTACT

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